
FACE

Numbers

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Question 1

Each of X alarms tolls at regular intervals. All of them toll together twelve times a day. No two alarms toll at equal intervals of time. If each alarm tolls after a whole number of minutes, what is the maximum possible value of X?

a. 14
b. 16
c. 18
d. 20

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Question 2

How many pairs of integers (x, y) exist such that the product of x, y and HCF (x, y) = 3000?

a. 5
b. 9
c. 10
d. 12

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Question 3

Let N be the largest number with exactly 3 factors that will perfectly divide 24! How many factors does (N + 4) have?

a. 12
b. 4
c. 6
d. 10

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Question 4

A positive number is divided by 100 to get a remainder thrice as the quotient. If the number is divisible by 11, how many such numbers are possible that are less than 100000?

a. 8
b. 5
c. 3
d. None of these

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Question 5

N! is completely divisible by 13^{52}. What is sum of the digits of the smallest such number N?

TCS

a. 17
b. 15
c. 16
d. 19

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Question 6

Find the greatest value of N for which 50! / 15N is an integer.

a. 16
b. 11
c. 12
d. 13

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Question 7

Find the remainder when $197^{34} + 251^{55}$ is divided by 28.

a. 1
b. 14
c. 0
d. 27

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Question 8

There are 1000 employees in a firm. All of them sign in a register having 1000 pages numbered from 1 to 1000. Employee number 1 signs on all pages. Employee number 2 signs on all pages whose page numbers are divisible by 2. Employee number 3 signs on all pages whose page numbers are divisible by 3. In this manner employee number N signs on all pages whose page numbers are divisible by N, for
$1 \leq N \leq 1000$. How many pages among the first 100 pages have exactly two signatures?

TCS

a. 30
b. 25
c. 27
d. 24

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Question 9

What is the remainder when $1(1!) + 2(2!) + \ldots + 10(10!)$ is divided by 35?

a. 1
b. 17
c. 18
d. 34

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Question 10

Find the 260th term of the series a, b, b, c, c, c, d, d, d, d, e, e, e, e, e, f, f, f, f, f, f,...

TCS

a. u
b. w
c. t
d. v

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Question 11

The HCF of two natural numbers is 33 and their sum is 528. Find the number of pairs of numbers satisfying the above condition.

a. 4
b. 5
c. 3
d. 6

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Question 12

Find the number of unique prime factors of 44444444.

a. 3
b. 4
c. 5
d. 6

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Question 13

The LCM and HCF of three numbers are 9044 and 4 respectively. lf two of the numbers are 68 and 28, find the least value of the third number. CTS

a. 76
b. 48
c. 96
d. 88

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Question 14

The product P of three positive integers is 9 times their sum, and one of the integers is the sum of the other two. Find the sum of all possible values of P.

a. 621
b. 702
c. 540
d. 336

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Question 15

Let a, b, c, d and e be distinct integers in ascending order such that (76 - a) (76 - b) (76- c) (76 - d) (76 - e) = 1127. What is a + b + c + d? TCS

a. 334
b. 274
c. 30
d. 136

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THANK YOU

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